

Design and Simulation of Sb₂Se₃-based Resonant Optical Limiters using Transfer Matrix Method

Surawut Wicharn¹, Chettanat Padungatthakij², Suebtarkul Suchat³, Nathan Hagen⁴, and Prathan Buranasiri^{2*}

¹ Department of Physics, Faculty of Science, Srinakharinwirot University, Bangkok, Thailand.

² Department of Physics, School of Science, King Mongkut's Institute of Technology Ladkrabang, Bangkok, Thailand.

³ Department of Physics, Faculty of Science and Technology, Phranakhon Rajabhat University, Bangkok, Thailand.

⁴ Department of Optical Engineering, Utsunomiya University, Tochigi, Japan.

* prathan.bu@kmitl.ac.th

Abstract:

We proposed an optical limiter designs which utilize antimony triselenide (Sb₂Se₃). Optical limiter is a type of device that allows light to transmitted through in one mode, but can be switched off to block transmission. Sb₂Se₃ is a phase change material which has low loss in the IR spectrum, high and stable refractive index contrast in both visible and IR spectra, and a high phase switching endurance of up to an order of 10⁶ cycles. Our optical limiters designs are for either 532 or 1064-nm usage; these two wavelengths have a distinctive design approaches, at 1064 nm Sb₂Se₃ has no absorptivity so it is suitable for a high efficiency optical limiters, but at 532 nm Sb₂Se₃ exhibits high absorptivity which complicated resonant design. Aside from the optical performance, we also investigate their optical endurance with computer simulations. (137 words)

Keywords: optical limiter, resonant cavity, Sb₂Se₃, phase change material (keywords < 7)

1. Introduction

Phase change material (PCM) can have different meaning depending on the topics. When discussing about energy storage, the term would refer to material that can store and release its thermal energy when changing its state of matter (latent heat), and the transition occurs between solid and liquid state [1]. However, in electronics and photonics, the term PCM refers to a solid-state material that changes its crystal structure, such as from amorphous to crystalline, when its temperature rise above transition temperature (T_{trans}). The change in crystalline structure leads to a change in its optical properties, i.e. complex refractive index [2]. Complex refractive index comprises of real refractive index (n) and imaginary extinction coefficient (k) parts, and the refractive index contrast between the two phases of PCM is an order of magnitude larger than contrast induced by applied electric field or light intensity (Kerr effect) [3, 4]. This property makes PCM ideal for photonic applications that require the toggle of light transmittivity, absorptivity, or reflectivity. Examples include optical limiters[4-10], optical switches[11-14], optical filters[15-17], optical diode[18], and absorbers[19-21].

Phase change material can be classified by its behavior after phase transition occurs. A volatile PCM will revert to its pre-transitioned phase if temperature drops below T_{trans} , a common volatile PCM is Vanadium dioxide (VO₂). But non-volatile PCM will remain in its transitioned phase even after the temperature drop below T_{trans} . To turn non-volatile PCM phase back to pre-transitioned phase, it needs to be heated to melting temperature and the quench rapidly. Examples of non-volatile PCMs are Ge₂Sb₂Te₅ (GST) and Ge-Sb-Se-Te (GSST). One of the first and most commonly studied non-volatile PCM is GST [22, 23]. It's a chalcogenide glass that exhibits large refractive index contrast between amorphous and crystalline phases. When compared to other non-volatile PCM, it has low T_{trans} at ~145 °C for cubical crystalline phase transition and changes to hexagonal crystalline phase at ~200 °C [4, 24]. Its drawback is the large extinction coefficient in visible and IR spectrum, which causes high optical loss from absorption. Another popular non-volatile PCM is GSST (Ge₂Sb₂Se_xTe_x), it is a derivative of GST is (GSST) that substitutes some of tellurium with selenium. It has a larger refractive index contrast in the visible spectrum like GST but has a lower extinction coefficient which helps with efficiency. But GSST has higher T_{trans} at around 242 °C [24]. In the past 10 years, VO₂ has gathered attention as a volatile PCM with T_{trans} close to room temperature. It undergoes metal-insulator transition (MIT) at T_{trans} of 68 °C and has a relatively high refractive index contrast starting from near-IR with small k in the insulating phase (low temperature). Its drawbacks are its large k in its metallic phase [23], noticeable thermal hysteresis [25]. Attempts have been made to reduce T_{trans} of VO₂ closer to room temperature, but these affect the switching endurance of the devices and reduce their refractive index contrast [26]. Another example of VO₂ usage aside from those mentioned above is a device such as a thermochromic window coatings that helps with temperature regulation inside a room by blocking infrared sunlight in the summer, but allows light to come in during winter to help with indoor heating [27-29].

Compared to other PCMs, antimony triselenide (Sb₂Se₃) is a relatively new material in the field of photonics, only one review paper exist prior to 2026 [30]. (recheck claims whether it's still true). The initial researches of Sb₂Se₃ are in solar energy applications [31], but it has recently gathered interest as a non-volatile

PCM for photonic applications [30] because of its near-zero extinction coefficients in the NIR spectrum and beyond, its extinction coefficient is also lower than GST in the visible spectrum for both amorphous and crystalline phases, it has a relative high and stable refractive index contrast from visible to IR spectrum, and has a fast switching cycle in the microsecond range [22, 32, 33]. Sb₂Se₃ crystallization temperature is between GST and GSST at around 186 °C [32]. Its phase switching endurance is also high, at 4000 cycles [22], and has also been experimentally demonstrated to be as high as 10⁶ to 10⁷ cycles with an optimized laser-induced crystallization and re-amorphization schemes [32, 33]. These properties make Sb₂Se₃ popular as a low-loss, reconfigurable optical switches or circuits that have been used in optical waveguide [11, 22, 34, 35]. But, to the best of our knowledge, Sb₂Se₃ application as optical limiter has never been explored before.

Optical limiter is a photonic device that protects detector or eyes from high intensity light, it allows light to be transmitted through during its “ON” mode but will block transmissions when switch to “OFF” mode. The device could be self-switching, i.e. undergoes phase transition via the absorption of light [7, 8], or it could switch its mode by external means, such as joule heating [16], or an absorption of light outside of its intended operation bandwidth [32]. The device could either be designed to operate in a broadband [4-6, 36] or single wavelength [7, 8, 37, 38]. An ideal optical limiter should maximize its transmittivity in ON but minimize in OFF mode, while minimize reflectivity and absorptivity in both modes. If it's self-activating, it should have just enough absorption to change its phase to minimize transmission loss in the ON mode. Optical limiter designs could be a thin-film planar or a nanostructured metasurface; planar designs are easier to design, optimize and fabricate, but metasurface designs offer more ways to optimize [8]. For a planar design, there are many ways to create high transmission such as 1) single or multi-layers antireflection (AR) designs and 2) Line-pass, long-pass, or short-pass filters [39]. All of these designs rely on resonant for AR property. But if the materials used have high extinction coefficients (k index), e.g. metals or some PCMs, a resonant condition will be disrupted due to energy loss correlated to the thickness of high- k index film. It can be restored by the use of asymmetrical resonant cavity [40], where such a design incorporates high k -index material placed between two mirror-like layers. The cavity designs have an effect of reducing the total thickness of PCM when compared to other designs, this is especially helpful in the case of PCMs because it reduces the total energy required for phase transitions. Another benefit of thinner total PCM used is the reduction of stress damage due to the contraction and expansion by crystallization and re-amorphization processes [17], negate partial phase transition from the thickness limit of the film and open up more laser optimization schemes for inducing phase transitions [32].

In this work, we investigate Sb₂Se₃-based optical limiters with a planar thin-film designs. The reference wavelengths are selected to be 1064 or 532 nm since its from Nd:YAG laser and its second harmonic which is commonly used in the lab. Another reason is because the two wavelengths show a different design approaches. At 1064 nm Sb₂Se₃ has near-zero absorptivity in both phases and the refractive index contrast is stable throughout IR spectrum. The goal of the 1064-nm designs is to create a broadband (IR) optical limiter with high transmissivity in the ON mode and reflected most light in the OFF mode. As for 532-nm designs, at this wavelength the k -indices of Sb₂Se₃ are at 1.09 and 2.24 for amorphous and crystalline phase, respectively [22]. This means a simple AR, or long/short-pass filters cannot be used, as described above. An asymmetrical resonant cavity design approach will be used for 532-nm design.

2. Theories

2.1 Optical Properties of Sb₂Se₃

The complex refractive index ($\tilde{n}$) of phase change material at any λ wavelength could be written as:

$$\tilde{n}_p(\lambda) = n_p(\lambda) + ik_p(\lambda) \quad (1)$$

where the real term n is the refractive index, the imaginary term k is extinction coefficient, and p indicates phase which for Sb₂Se₃ can either be amorphous (a) or crystalline (c). Dispersive models, such as Tauc-Lorentz or multi-oscillator fits are often used to represent the measured n_p and k_p over the spectral range of interest. (need this part?). In this work, since all the materials dataset used have measurement interval of 2-5 nm, a linear interpolation fitting should be sufficient. The complex refractive index of Sb₂Se₃ from 300-1700 nm are shown in Fig. 1 using dataset from [22]. The blue line represents real refractive index (n), and the red line represents imaginary extinction coefficient (k). In amorphous phase (Fig. 1a), n -index increases from around 2.5 at 300 nm to a peak of around 4.0 at 600 nm and gradually decreases to a stable 3.3 at 1200 nm, indicating a strong dispersion near the band edge of visible spectrum and a weaker dispersion in the near-IR spectrum. The k -index started from around 2.1 at 300 nm and dropped to zero at around 820 nm. In the crystalline phase (Fig. 1b), the n -index started around the same value as in amorphous phase but reached a peak of around 5.0 at 650 nm, it is then dropped to a stable of 4.0 in the IR spectrum. The k -index of crystalline phase peaked at around 3.1 in the near-UV spectrum and rapidly falls to zero above ~900 nm. So, the refractive index contrast $\Delta n = n_c - n_a$ is large and stable in the visible through IR spectra, and extinction coefficients is very large in the near-UV but falls to zero around near-IR spectrum for both phases, with crystalline Sb₂Se₃ having higher absorptivity in the visible spectrum.

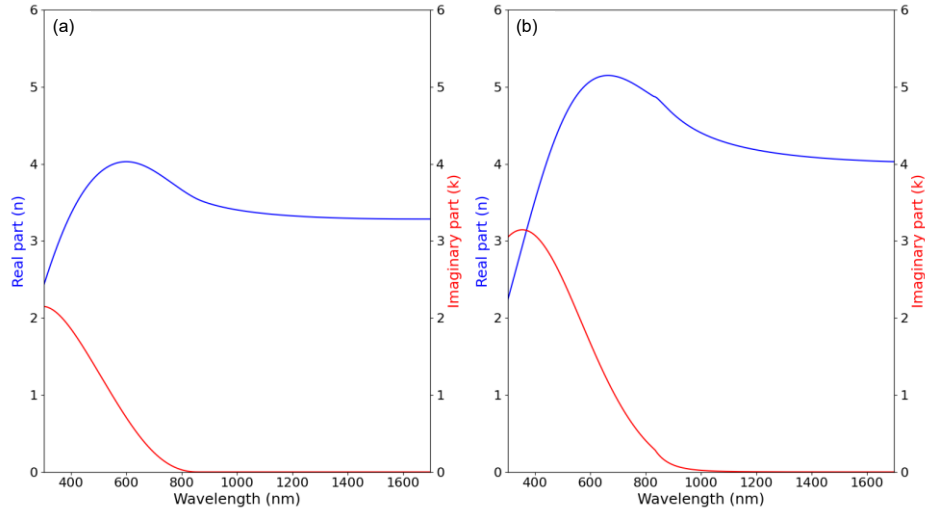


Fig. 1. Complex refractive indices of Sb_2Se_3 in its (a) amorphous, and (b) crystalline phase. The plotted data are taken from [22].

2.2 Single-layer antireflection design

In designing optical limiters, we will consider its optical performance in the ON mode first, and then we can consider its transmission contrast when switched to OFF mode. Most types of optical devices will either be optimized for antireflection (AR) or high reflection properties but since optical limiters only utilize transmission and absorption properties, minimizing reflection is beneficial (absorptivity is utilized for phase transition). The simplest thin-film device consists of a single film layer with no absorption (k -index = 0) deposited on a substrate. A single-layer device reflects light at two interfaces as shown in Fig. 2: the first interface is between thin film and substrate (R_1), and the second interface is between air and thin film (R_2). Antireflection occurs when destructive interferences of both R_1 and R_2 with the same intensity but a phase difference of π radian, or optical path length of $\lambda/2$, occurs. And since R_1 travelled through the film layer twice (down and up), the film optical thickness only needs to be half of that, $\lambda/4$ optical-path thick which is also known as quarter-wave optical thickness (QWOT).

Optical thickness (OT) is the film layer optical path length of light at wavelength λ , which can be defined as:

$$OT(\lambda) = n(\lambda)t, \quad (2)$$

where n is the refractive index of the film at λ , and t is the physical thickness of the film. For simplicity reason, λ is equal to the central wavelength (λ_0), i.e. reference wavelength and will be omitted in the following equations. For a single-layer AR device, the thin-film physical thickness (t) needs to be equal to:

$$t = \frac{OT}{n} = \frac{\lambda}{4n}. \quad (3)$$

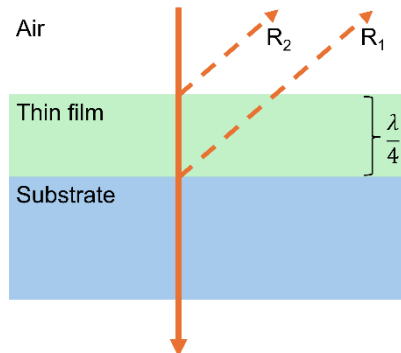


Fig. 2. Light paths in a device made of single-layer film. R_1 is the reflected light at the film-substrate interface, and R_2 is the reflected light at the air-film interface. Antireflection occurs when the phase difference between R_1 and R_2 is π radian. The angles in this figure are exaggerated, both incident and reflected angles are 0°

Another way to find thickness required for AR condition is to consider the reflectivity. At the air-film interface (R_2), the reflectivity (r) can be described as:

$$r = \frac{n_a - n_f}{n_a + n_f}, \quad (3)$$

where n_a is the refractive index of air, and n_f is the refractive index of the film. But since reflection also occurs in the R_1 interface (film-substrate), effective reflectance index (n_r) must be used in place of n_f in Eq.3. Effective reflectance index is the combined refractive index of film and substrate [39],

$$n_r = n_f \frac{(n_s + n_f) + (n_s - n_f)e^{-i2\theta}}{(n_s + n_f) - (n_s - n_f)e^{-i2\theta}}. \quad (4)$$

where n_s is the substrate refractive index, i denotes imaginary number, and θ is the phase thickness which is related to optical thickness (OT) of the film by

$$\theta = k_0 OT, \quad (5)$$

where k_0 is the angular wavenumber ($k_0 = 2\pi/\lambda$). By replacing n_f with n_r Eq. 3 then become

$$r = \frac{n_a - n_r}{n_a + n_r}. \quad (6)$$

Antireflection property is achieved when $r = 0$ or $n_a = n_r$. This condition occurs only when the phase thickness (θ) in Eq.4 is in multiple of $\pi/2$ radian, or QWOT:

$$\theta = Z \frac{\pi}{2}, \quad (7)$$

where Z can be any integer. If Z is even number, we can simplify Eq.4 using Euler's identity ($e^{-i\pi} = -1$) to:

$$n_r = n_f \frac{(n_s + n_f) - (n_s - n_f)}{(n_s + n_f) + (n_s - n_f)} = \frac{n_f^2}{n_s}. \quad (8)$$

Antireflection condition occurs when $n_r = n_a$ or:

$$n_a n_s = n_f^2, \quad (9)$$

another condition to consider is the relative refractive index between air, film, and substrate such that $n_{air} < n_f < n_{substrate}$. Since reflected light shift its phase by π radian when travelled from lower to high refractive index medium. The R_2 light shift its phase by π radian since n_{air} is the lowest in most cases, so R_1 must have it phase shifted by the same amount. These two conditions does place a limitation on material refractive index selection, since there exist only one possible answer for each substrate to create a perfect antireflection condition. The QWOT ($\lambda/4$) is only usable when film refractive index is less than substrate, which in this work is silica (SiO_2). Sb_2Se_3 has refractive index greater than silica at both 532 and 1064 nm wavelengths, R_1 light travelled 2QWOT = π radian further than R_2 but R_2 also has it phase shifted by π from the reflection, so the overall reflection actually increased. In this case, the film needs to be $\lambda/2$ or half-wave optical thick (HWOT) so that R_1 would travel 2HWOT = 2π radian further than R_2 . As an example, if we use 3 mm-thick SiO_2 substrate with a refractive index of 1.46 at 1064 nm [41] n_f then needs to be 1.21 to create a complete antireflection design. The value 1.21 is lower than any commonly available material and the material with the closest refractive index is MgF_2 at 1.38 [42], this difference cause the reflectivity to be higher than an ideal film as can be seen in Fig. 3 (dotted vs dashed line). In the case of Sb_2Se_3 (solid line), its thickness is HWOT as explained above. The reason an ideal film still has some reflectivity even though has a perfect AR condition is because we modelled the reflection from substrate-air interface on the bottom.

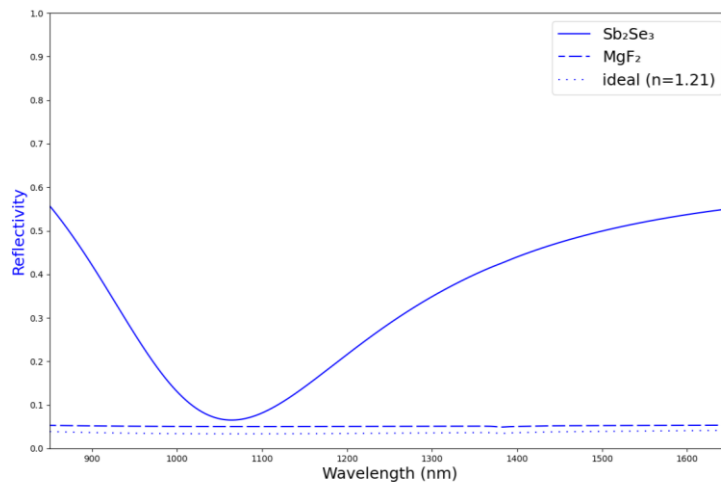


Fig. 3. Reflectivity of single-layer thin film design on a 3-mm thick Silica substrate. An ideal film with $n_f = 1.21$ (dotted line) has reflection = 3.4%, MgF_2 with $n_{MgF_2} = 1.38$ (dashed line) has reflection = 5%, and Sb_2Se_3 (solid line) has reflection = 6.5%.

These models are only true substrate is infinitely thick and there is no film on the backside of the substrate. Otherwise, reflectance must be used, it can be described as a ratio between reflected and incident power. (true?)

2.3 Multilayer antireflection design

A two-layer device with QWOT on each layer (Q/Q design) can also create a perfect antireflection condition. The reflectivity (r) from Eq.6 can be expanded to:

$$r = \frac{n_a - n_{r2}}{n_a + n_{r2}}, \quad (10)$$

$$r = \frac{n_a - \left(\frac{n_{f2}^2}{n_{r1}}\right)}{n_a + \left(\frac{n_{f2}^2}{n_{r1}}\right)}, \quad (11)$$

$$r = \frac{n_a - \left(\frac{n_{f2}^2}{n_{f1}^2/n_s}\right)}{n_a + \left(\frac{n_{f2}^2}{n_{f1}^2/n_s}\right)}, \quad (12)$$

where the film numbering (n_f) starts from substrate side and increases toward air ($n_s / n_{f1} / n_{f2} / n_a$), n_{r2} is the effective reflectance index of the device [39], and n_{r1} is the effective reflectance index of only n_s and n_1 . Antireflection condition ($r = 0$) occurs when:

$$n_a = \frac{n_{f2}^2}{n_{f1}^2/n_s}, \quad (13)$$

or:

$$\frac{n_a}{n_s} = \frac{n_{f2}^2}{n_{f1}^2}. \quad (14)$$

This means the refractive index value of top film must be less than bottom film ($n_{f2} < n_{f1}$). As an example, at 1064 nm if we use the same SiO_2 as substrate and Sb_2Se_3 as bottom film ($n_{f1}=3.37$), the refractive index of n_{f2} is calculated to 2.78. A commonly used material with the closest refractive index is TiO_2 with $n_{f1}=2.31$ [43], so a complete antireflection cannot be achieved. As can be seen in Fig. 4, when compared to a single-layer design, two-layer design allows for wider materials selection, has lower reflectivity, and broader operating wavelength.

Another way to create a perfect antireflection condition is to fix the two materials refractive index and vary the thickness instead, as described in chapter 5.2 of [39]. But this narrows the antireflection bandwidth even further.

A three-layer design (Q/H/Q) has a HWOT layer placed in the middle. The refractive index values of each layer required are such that $n_s < n_{f1} < n_{f2}$ and $n_{f2} > n_{f3} > n_a$. The middle layer (n_{f2}) is an absentee layer when incident wavelength (λ) is the same as reference wavelength (λ_0), but it acts as compensating layer for restoring AR condition for a scenarios where λ differs from λ_0 , a more detailed explanation can be found in chapter 5.4 of [39]. The reflectivity from Eq.6 is equal to:

$$r = \frac{n_a - n_{r3}}{n_a + n_{r3}} = \frac{n_a - \left(\frac{n_{f3}^2}{n_{r2}}\right)}{n_a + \left(\frac{n_{f3}^2}{n_{r2}}\right)}, \quad (15)$$

because the middle film layer (n_{f2}) thickness is HWOT, it n_{f2} behaves as an absentee layer [39, 44]. So $n_{r2} = n_{r1}$ and Eq. 15 become:

$$r = \frac{n_a - \left(\frac{n_{f3}^2}{n_{r1}}\right)}{n_a + \left(\frac{n_{f3}^2}{n_{r1}}\right)} = \frac{n_a - \left(\frac{n_{f3}^2}{n_{f1}^2/n_s}\right)}{n_a + \left(\frac{n_{f3}^2}{n_{f1}^2/n_s}\right)}. \quad (16)$$

A three-layer antireflection condition occurs when:

$$\frac{n_a}{n_s} = \frac{n_{f3}^2}{n_{f1}^2}. \quad (17)$$

The ratio between top and bottom film index is the same as the two-layer (Q/Q) design, which means a complete AR condition also cannot be achieved. The benefit of adding HWOT middle layer (n_{f2}) is it contribution to a broader operational bandwidth [44]. Its reflection at wavelength further from λ_0 , such as at 1400 nm and above, is lower than in a single or two-layer designs (Fig. 4). But **instead of using Sb_2Se_3 as the bottom film layer (n_{f1}), it can be moved to the middle layer (n_{f2})**, the selection of n_{f1} and n_{f3} could then be more practicable. For example, at 1064 nm, the bottom layer could be Si_3N_4 ($n_{f1}=2.00$ [45]). Using equation 17, the top layer (n_{f3}) then needs to be 1.65, Al_2O_3 has a closest refractive index of 1.62 [43].

This is illustrated in Fig. 4 where the reflection of the single-layer, two-layer, and three-layer designs, all of which has Sb_2Se_3 placed on top of the substrate ($n_{f1} = \text{Sb}_2\text{Se}_3$), are around 6.7%. But when we placed Sb_2Se_3 in the middle ($n_{f2} = \text{Sb}_2\text{Se}_3$) reflection dropped to 3.5% at a broader wavelength, this could be explained by the selections of n_{f1} and n_{f3} refractive indices are closer to ideal values.

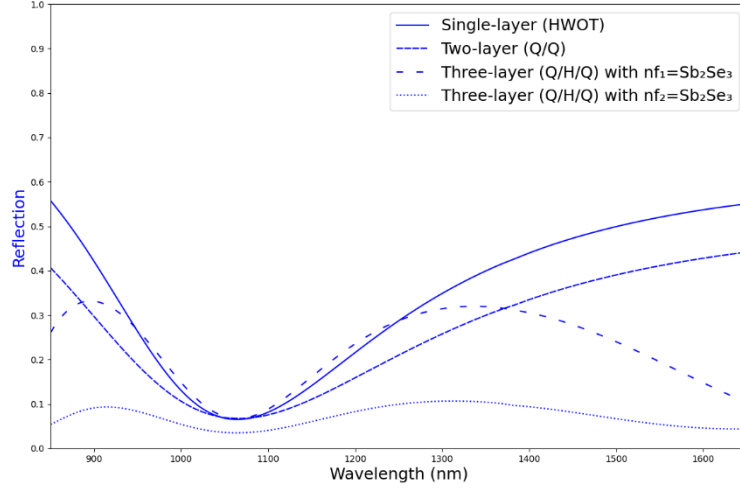


Fig. 4. Reflection comparison of a single-layer, two-layer, and three-layer antireflection designs. The single-layer design is the same as in Fig. 3. The two-layer and three-layer with $n_{f1} = \text{Sb}_2\text{Se}_3$ designs have the same reflection of 6.7% at 1064 nm, comparable to a single-layer design. Three-layer design with $n_{f2} = \text{Sb}_2\text{Se}_3$ with HWOT has lower reflection at 3.5%, around the same the ideal film single-layer design in Fig. 2.

2.4 Fabry–Pérot cavity design

In all the previous designs, we haven't considered its behaviour after phase transition. If we calculate the previous designs in Fig.4 again but with refractive index of Sb_2Se_3 in crystalline phase instead of amorphous phase, the design with the highest reflection contrast is a single-layer design at 55.0% (Fig. 5a). As for three-layer design with Sb_2Se_3 in the middle layer, even though it shows the best performance in the amorphous phase, its reflection contrast is the lowest at only 0.5% (Fig. 5b). Which means that aside from low reflection, a reflection profile with a sharp increase in reflection when λ move beyond the initial λ_0 is also required for optical limiters.

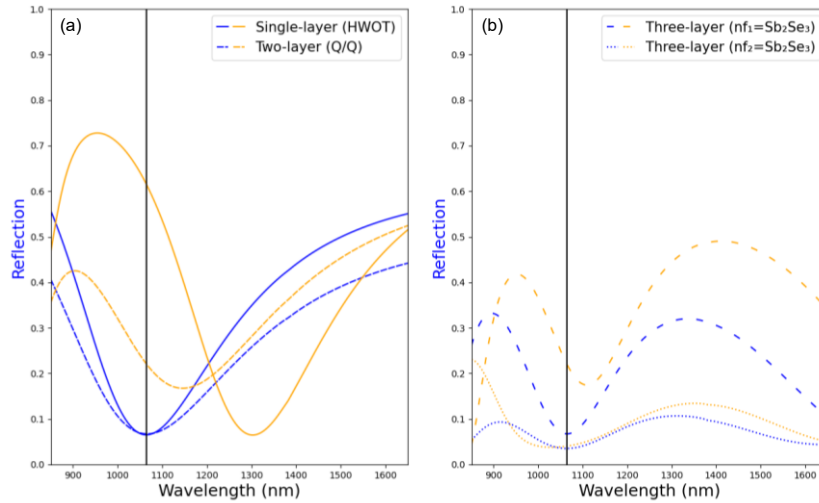


Fig. 5. Reflection comparison of previous designs before (blue) and after (orange) phase transition. (a) shows a single and two-layer designs. (b) shows a three-layer designs with Sb_2Se_3 either on the bottom or middle layer. A single-layer design shows the highest reflection contrast when undergoes phase transition. Black vertical line indicates $\lambda_0 = 1064$ nm.

A Fabry–Pérot cavity design acts as edge filters which has high reflection around the reference wavelength (λ_0), but has high transmission beyond the edges. The design can either be $(\frac{H}{2}L\frac{H}{2})$ or $(\frac{L}{2}H\frac{L}{2})$ symmetrical unit cell(s) for high-pass or low-pass design, respectively, where H and L is material with high and low refractive index contrast, respectively, and H or L are QWOT films and $\frac{1}{2}$ denotes a $\frac{\text{QWOT}}{2}$ films. Since Fabry–Pérot cavity shows high reflection at λ_0 , and we design our optical limiters to operate at 1064 nm, the reference wavelength must be move to a shorter (high-pass) or longer (low-pass) wavelength. Shorter λ_0 is preferred since

it allows for thinner device so a high-pass ($\frac{H}{2} L \frac{H}{2}$) design will be used. The spectral width of high reflection region is determined by the difference between H and L refractive indices, the higher the difference the larger the area. The sharpness of the edges is determined by the number of unit cell. Figure 6 is an example designs with ($\frac{H}{2} L \frac{H}{2}$) unit cell(s), Sb_2Se_3 as H and MgF_2 as L layer, and $\lambda_0 = 750$ nm. Figure 6a is the comparison between different number of unit cells, designs with higher cell numbers have a sharper edge. It is interesting to note that design with 8-unit cells has sharper edge than 4-unit cells design, but the reflection at around 1350 and 1600 nm are higher, so higher unit cells doesn't necessarily have better performance. Figure 6b shows a comparison of amorphous and crystalline Sb_2Se_3 , the reflection of amorphous and crystalline phase is 6.5% and 94.1%, respectively, so the reflection contrast at 1064 nm is 87.6% which is the highest of all designs shown so far. Another point to note is that the reflection in amorphous phase is around the same as other design with Sb_2Se_3 on the bottom (directly on top of the SiO_2 substrate).

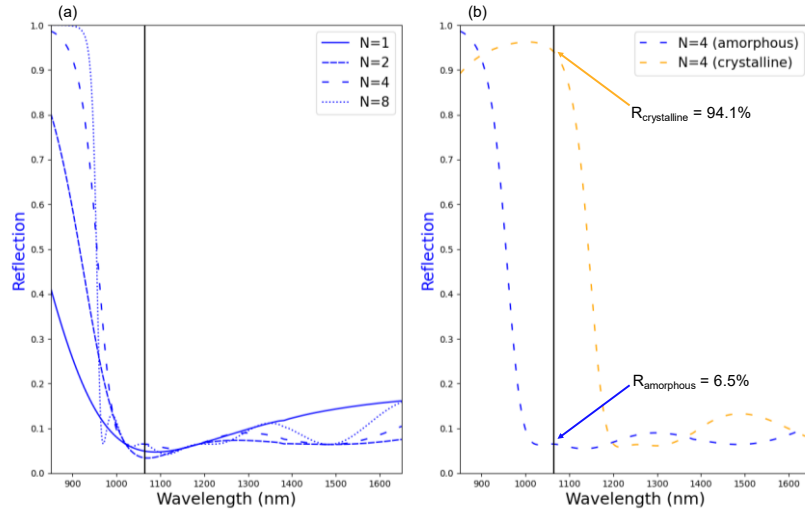


Fig. 6. Reflection comparison of Fabry-Pérot cavity ($\frac{H}{2} L \frac{H}{2}$) designs with (a) varying number of unit cells and (b) design with 4-unit cells before and after phase transition. Black vertical line wavelength is 1064 nm, the interested optical limiter operating wavelength. All designs have reference wavelength $\lambda_0 = 750$ nm.

Another planar thin-film design is a resonant cavity design, which can be used to enhance the optical properties of the middle layer (PCM in this work) [40]. The cavity consists of PCM layer in the middle and a distributed Bragg reflectors on each side of the PCM, as shown in Fig. 3. The two reflectors function as mirrors that cause light to cycles through the central PCM layer multiple times, this effectively reduced the required thickness of PCM. This benefit is especially useful for PCM with high T_{trans} such as Sb_2Se_3 because it lowers the energy requirement to initiate phase transition, and negates damage caused by partial phase transition due to PCM being too thick [17, 32]. The mirror structure can be described as a form of unit cell (U) composed of high (H) and low (L) refractive index materials with QWOT. The difference between H and L index directly correlated to the enhancement. In this work, we denoted unit cell as $U=[H/2 \ L \ H/2]$, where 1/2 indicates a film with half of QWOT or $\lambda/8$. For example, a symmetrical cavity with $U=1$ on each side of the central (C) PCM layer can be written as:

$$\frac{H}{2} U^1 C U^1 \frac{H}{2}, \quad (18)$$

$$\frac{H}{2} \left[\frac{H}{2} L \frac{H}{2} \right] C \left[\frac{H}{2} L \frac{H}{2} \right] \frac{H}{2}, \quad (19)$$

The limitations of this design is its narrow operation range, and the difficulty in fabrication process.

2.5 Antireflection design in high-loss film (532)

In a single-layer design of Fig.2, zero reflection happens when the phases of R_1 and R_2 is opposite but their amplitudes are the same. If the film has any k-index, such as in the case of Sb_2Se_3 , some of the light will be absorb and R_1 will have lower amplitude than R_2 . This effect is especially magnified in resonant cavity design. One way to visualize this is to plot reflectivity (r) in a contour graph. Figure 4 illustrates the effect material with k-index has on resonant loss. The two designs are optimized for resonant at 1064 nm wavelength, they have $U=2$ on each side of the central layer. Refractive index of H and L films are the same for both designs, 2.45 and 1.63, respectively. These two designs assume substrate to have the same refractive index as air. The central cavity layer

in Fig.5a is equal to $C=3.96+0i$. For Fig.5b central layer is the same material except it has k-index term, $C=3.96+1.09i$.

3. Methods

3.1 Optical limiter design

In this work, we use transfer matrix method (TMM) of Fresnel equation to calculate the transmittance, absorptance, and reflectance of devices. In some simulation works, the substrate index can consider to be equal to air ($n_{sub} = n_{air} = 1.0$) to simplify the design and allows for a more widely applicable analysis methods, this could be done in reality by applying antireflection coating on the back side of the substrate [40]. But in this work, we choose to use 3-mm thick silica (SiO_2) as substrate with refractive index data taken from [46], its refractive index is around 1.47 at both 1064 and 532 nm wavelengths. The H layers are Si [47], TiO_2 [43] and Si_3N_4 [45]. The L layers are Al_2O_3 [43], SiO_2 and MgF_2 [42]. All the above mentioned material refractive indices can be found on refractiveindex.info website [48], but since antimony triselenide (Sb_2Se_3) is a relatively new material in the field of optics, its refractive index couldn't be found there. We use dataset from [22], as shown in Fig. 1. The refractive index of air is set $n_a = 1.0$ across the spectra. The incident angle is 0° and light is transverse electric (TE) polarized for all calculations.

Our optical limiters are designed will either be designed for 1064 or 532 nm wavelengths. For 1064-nm devices, the designs are single-layer, multi-layer, and edge-filter. The goals are to design broadband optical limiter with large transmission contrast (ΔT) between the two modes. **Because of the low absorption caused by near-zero k-index in both phases at this wavelength, the devices cannot be self-activated.** Since Sb_2Se_3 have high k-index at 532-nm wavelength, $k_a = 1.09$ and $k_c = 2.24$, we will compare regular design approaches (e.g., single or multi-layer designs) to asymmetrical cavity design. And because of the high absorptivity, the devices can be self-activated type at this wavelength.

3.2 Simulation of endurance and optical stability under phase switching cycles

Experimental studies have demonstrated that Sb_2Se_3 can have phase switching cycles with minimal degradation to its optical contrast up, **and maintain a low-loss, reversible modulation** in an order of 10^4 [cite] and 10^6 cycles [32]. Even so, Sb_2Se_3 still exhibits slight complex refractive index drift due to structural relaxation and cumulative thermal damage. Accounting for these variations is essential for ensuring long-term reliability and precision in high-endurance photonic applications [32, 33]. To account for this, the complex refractive index ($\tilde{n}$) can be rewritten from Eq. 1 as:

$$\tilde{n}_p(N) = n_p(N) + ik_p(N), \quad (\text{XXX})$$

where N is the number of phase switching cycle.

The real part of the refractive index (n) is described by an exponential relaxation towards a saturated value such that:

$$n_p(N) = n_{p,\infty} - (n_{p,\infty} - n_{p,0})\exp^{-\alpha_p N}, \quad (\text{XXA})$$

where $n_{p,\infty}$ is the **asymptotic index** after many cycles, $n_{p,0}$ is the refractive index at the first cycle (**obtained from ellipsometry or literature data**), and α_p is the drift coefficient per cycle of phase p [22, 49].

Meanwhile, the imaginary part (k) can be approximated from a linear growth as:

$$k_p(N) = k_{p,0}(1 + \beta_p N), \quad (\text{XXB})$$

where $k_{p,0}$ is the initial **loss** and β_p is the loss growth coefficient. The parameter β_p is the relative loss per cycle of phase p [32, 50]. These models from Eq. (XXA) and (XXB) are adopted in this work, they are consistent with experimental trends and with general PCM endurance theory, providing a practical way to assess contrast degradation and threshold drift of PCM-based optical limiters over large cycles [22, 32, 49-51]. A more detailed derivation these endurance models are shown in **Supplementary Material**.

The endurance of optical limiters can be quantified using the following metrics: 1) transmitting contrast degradation, and 2) insertion loss.

Transmitting Contrast Degradation:

Degradation is defined as the reduction in transmission contrast after N cycles, as in:

$$D(N) = \frac{\Delta T(N) - \Delta T_0}{\Delta T_0}, \quad (\text{XX})$$

where transmission contrast ΔT_0 is the contrast of optical device between amorphous (ON) and crystalline (OFF) phases initially (cycle = 0). Transmission contrast (ΔT) can be defined as: **$(T_a - T_c \text{ or } T_c - T_a)$**

$$\Delta T(N) = T_a(N) - T_c(N). \quad (\text{XX})$$

Insertion loss:

The insertion loss of the device at cycle N compares it transmittance to a reference state of cycle 0 [52], such that: **$T_c(1)$ or $T_c(0)$?**

$$IL_{dB}(N) = 10 \log \left[\frac{T_c(N)}{T_c(1)} \right]. \quad (XX)$$

A growing $IL(N)$ indicates increasing loss, typically driven by the linear growth of the imaginary index $k_p(N)$ and by detuning of the resonance due to drift of $n_p(N)$.

4. Results

4.1 1064-nm optical limiters

Antimony triselenide shows little to no absorption in the infrared spectra, so an ideal design could have 100% transmission in the ON mode and 0% in the OFF mode. The transmission, absorption, reflection, Sb_2Se_3 thickness, overall device thickness and the number of layers needed for each previously discussed design are show in Table 1. Fabry-Perot designs have better transmission contrast (ΔT)

Table 1. Designs of 1064-nm optical limiter. Transmission is calculated at $\lambda = 1064$ nm. The Fabry-Perot cavity designs have $\lambda_0 = 750$ nm.

Design	Layout	Number of layers	SbSe used (nm)	Device size (nm)	T (ON) (%)	T (OFF) (%)	ΔT (%)
Single-layer	SbSe	1	157.9	157.9	93.5	37.2	56.3
Two-layer	SbSe/TiO	2	78.9	194.1	93.3	76.9	16.4
Three-layer	SbSe/MgF/TiO	3	78.9	580.3	93.3	76.9	16.4
Three-layer	SiN/SbSe/AlO	3	157.9	455.7	96.6	83.3	13.3
Fabry-Perot	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^1$	3	44.1	179.9	95.1	65.4	29.7
Fabry-Perot	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^2$	5	88.1	359.8	96.6	27.6	69.0
Fabry-Perot	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^4$	9	176.3	719.6	93.5	2.8	90.7
Fabry-Perot	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^5$	11	220.3	899.5	95.3	0.8	94.5
Fabry-Perot	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^6$	13	264.4	1,079.4	96.5	0.3	96.2
Fabry-Perot	$(\frac{\text{SbSe}}{2}/\text{MgF}/\frac{\text{SbSe}}{2})^8$	17	352.5	1,439.3	93.6	0.0	93.6

5. Discussions

For a 1064-nm optical limiters, a simple QWOT-design already satisfy anti-reflection conditions, it's also the smallest design. For a broader operating range in the IR spectrum, the XXX design has...

For a self-activating optical limiter at 532 nm, resonance can be restored with asymmetrical designs. Asymmetrical cavity designs required thinner Sb_2Se_3 . Which should prevent partial phase-transition described in [32]. The question then becomes 'at what threshold of laser intensity will the optical limiter be turned off? If the threshold is low, then the device needs to have more absorptivity in the ON mode. But if the threshold is high, then the absorptivity in the ON mode does not need to be as high. Its absorptivity in the OFF mode is also important; it ideally should be as high as possible since it need to be re-amorphized at an even higher temperature to return the device from OFF to ON mode. Would a volatile-type PCM be more suitable for self-activating optical limiter?

Another metric that we didn't show here are the tolerance of optical performances at varying incident angles. A high-tolerance devices would allow for broader applications.

Finite Element Method (FEM) [6]

The Fabry-Perot designs have PCM separated into many layers, which could complicated phase switching methods. If we use laser as switching method, the phase transition of each Sb_2Se_3 layer is not simultaneous, such that the transmission contrast will increase in a gradual sense, this effect can be seen in Sarangan, et al. [4]'s work, their self-activating optical limiter use $(\frac{H}{2}L\frac{H}{2})^4$ unit cells design but use GST as PCM instead of Sb_2Se_3 , their experimental result shows that the switching of GST is done layer-by-layer in a sequence determined by the changing optical field distribution and the thermal crosstalk between layers.

Our work still lacks thermal analysis which will determine the laser phase switching schemes for both 532 and 1064 nm design.

Note

6. Conclusions

7. Acknowledgement

8. References

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